Assignment No : 1

Q.1] Print an nxn Star Square

→

```
class Demo{
    Public Static void main
        (String [] args){

        For (int i=1; i<=4; i++){
            For (int j=1; j<=4; j++){
                System.out.Print("*");
            }
            System.out.println();
        }
    }
}
```

* * * *
* * * *
* * * *
* * * *

~~i=1~~ i=2 i=3 i=4 ~~i=5~~

$\boxed{i=1 \rightarrow i<=4 \rightarrow True}$ } Outer loop

1<=4 → T
↓
~~j=1~~
~~i<=4 → T~~

j = $\boxed{1}$ → j<=4 → False
↓
True
Sop("*") → $\boxed{j++}$

j = $\boxed{2}$ → j<=4
↓
True
Sop("*") → $\boxed{j++}$

j = $\boxed{3}$ → j<=4
↓
True
Sop("*") → $\boxed{j++}$

j = $\boxed{4}$ → j<=4
↓
True
Sop("*") → $\boxed{j++}$

j = $\boxed{5}$ → j<=4
↓
False Sopln()

आतल्या loop मधून बाहेर पडणार.

i++

* * * *

$\boxed{i = 2 \rightarrow i <= 4 \rightarrow True}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

* * * *
*

$j = \boxed{2} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

* * * *
* *

$j = \boxed{3} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

* * * *
* * *

$j = \boxed{4} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

* * * *
* * * *

$j = \boxed{5} \rightarrow j <= 4$
↓
$\boxed{False}$ sopln()

inner loop मधून बाहेर पडणार आणि i increment होणार.

$\boxed{i = 3 \rightarrow i <= 4 \rightarrow True}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

* * * *
* * * *
*

$j = \boxed{2} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

* * * *
* * * *
* *

$j = \boxed{3} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

* * * *
* * * *
* * *

$j = \boxed{4} \rightarrow j <= 4$
↓
True
sop("*") → $\boxed{j++}$

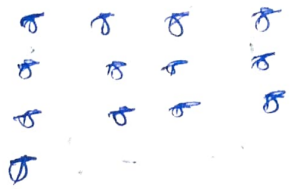
* * * *
* * * *
* * * *

$j = \boxed{5} \rightarrow j <= 4$
↓
$\boxed{False}$ sopln()

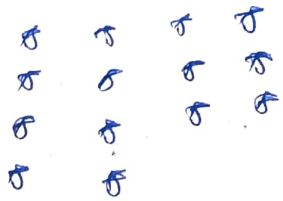
inner loop मधून बाहेर पडणार आणि i increment होणार.

$\boxed{i=4 \rightarrow i<=4 \rightarrow True}$

$j = \boxed{1} \rightarrow j<=4$
↓
True
Sop("*") → $\boxed{j++}$

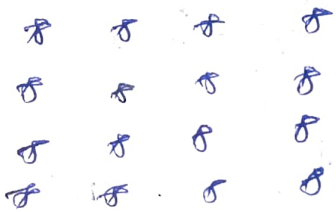


$j = \boxed{2} \rightarrow j<=4$
↓
True
Sop("*") → $\boxed{j++}$



$j = \boxed{3} \rightarrow j<=4$
↓
True
Sop("*") → $\boxed{j++}$

$j = \boxed{4} \rightarrow j<=4$
↓
True
Sop("*") → $\boxed{j++}$



$j = \boxed{5} \rightarrow j<=4$
↓
$\boxed{False}$ Sopln()

inner loop च्या बाहेर पडणार आणि i चे increment होणार.

$\boxed{i=5 \rightarrow i<=4 \rightarrow False}$

outer loop पण false आला त्यामुळे ~~दोन्ही~~ च्या loop च्या पण बाहेर पडणार.

∴ Hence code executed successfully.

Q.3] Print the Following Patten

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

→ class Demo {

```
    Public static void main(String[] args){
        For (int i = 1; i<=4 ; i ++){
            For (int j = 1 ; j<=4 ; j ++){
                System.out.Print (j);
            }
            System.out.println();
        }
    }
}
```

$\boxed{i = 1 \rightarrow i <= 4 \rightarrow True}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
Sop(1) → $\boxed{j++}$

1

$j = \boxed{2} \rightarrow j <= 4$
↓
True
sop(2) → $\boxed{j++}$

1 2

$j = \boxed{3} \rightarrow j <= 4$
↓
True
Sop(3) → $\boxed{j++}$

1 2 3

$j = \boxed{4} \rightarrow j <= 4$
↓
True
sop(4) → $\boxed{j++}$

1 2 3 4

$j = \boxed{5} \rightarrow j <= 4$
↓
False
↓
Sopln();

} inner loop मधून बाहेर पडणार आणि next line ला येऊन i increment होणार.

$\boxed{i = 2 \rightarrow i <= 4 \rightarrow True}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
Sop(j) → $\boxed{j++}$

1 2 3 4
1

$j = \boxed{2} \rightarrow j <= 4$
↓
True
Sop(2) → $\boxed{j++}$

1 2 3 4
1 2

$j = \boxed{3} \rightarrow j <= 4$
↓
True
Sop(3) → $\boxed{j++}$

1 2 3 4
1 2 3

$j = \boxed{4} \rightarrow j <= 4$
↓
True
Sop(4) → j++

1 2 3 4
1 2 3 4

$j = 5 \rightarrow j <= 4$
$\boxed{False}$
↓
Sopln();

inner loop च्या बाहेर येऊन i increment करणार.

$\boxed{i = 3 \rightarrow i <= 4 \rightarrow True}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
Sop(j) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1

$j = \boxed{2} \rightarrow j <= 4$
↓
True
Sop(2) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1 2

$j = \boxed{3} \rightarrow j <= 4$
↓
True
Sop(3) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1 2 3

$j = \boxed{4} \rightarrow j <= 4$
↓
True
Sop(4) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1 2 3 4

$j = \boxed{5} \rightarrow j <= 4$
↓
$\boxed{False}$ Sopln()

inner loop च्या बाहेर येऊन i increment होणार.

$\boxed{i = 4 \rightarrow i <= 4 \rightarrow \text{True}}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
sop(1) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1 2 3 4
1

$j = \boxed{2} \rightarrow j <= 4$
↓
True
sop(2) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1 2 3 4
1 2

$j = \boxed{3} \rightarrow j <= 4$
↓
True
sop(3) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3

$j = \boxed{4} \rightarrow j <= 4$
↓
True
sop(4) → $\boxed{j++}$

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

$j = \boxed{5} \rightarrow j <= 4$
↓
False sopln();

inner loop च्या बाहेर येणार
i increment होणार.

$\boxed{i = 5 \rightarrow i <= 4 \rightarrow \text{False}}$

outer loop ची घेण condition False
- loop च्या बाहेर पडणार.

∴ Hence code executed successfully

Yenjoyy!!

Q.2] Print the following Pattern

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

→

```
class Demo{
    public static void main(String [] args){
        for(int i=1; i<=4; i++){
            for(int j=1; j<=4; j++){
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

$\boxed{i=1 \rightarrow i<=4 \rightarrow True}$ → आपण inner loop मध्ये i print करणार आहोत.

$j=\boxed{1} \rightarrow j<=4$
↓
True
sop(1) → $\boxed{j++}$

1

$j=\boxed{2} \rightarrow j<=4$
↓
True
sop(1) → $\boxed{j++}$

1 1

$j=\boxed{3} \rightarrow j<=4$
↓
True
sop(1) → $\boxed{j++}$

1 1 1

$j=\boxed{4} \rightarrow j<=4$
↓
True
sop(1) → $\boxed{j++}$

1 1 1 1

$j=\boxed{5} \rightarrow j<=4$
↓
$\boxed{False}$
↓
sopln()

inner condition False झाली त्यामुळे बाहेर पडणार, i increment होणार.

$\boxed{i = 2 \rightarrow i <= 4 \rightarrow \text{True}}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
Sop(2) → $\boxed{j++}$

1 1 1 1
2

$j = \boxed{2} \rightarrow j <= 4$
↓
True
Sop(2) → $\boxed{j++}$

1 1 1 1
2 2

$j = \boxed{3} \rightarrow j <= 4$
↓
True
Sop(2) → $\boxed{j++}$

1 1 1 1
2 2 2

$j = \boxed{4} \rightarrow j <= 4$
↓
True
Sop(2) → $\boxed{j++}$

1 1 1 1
2 2 2 2

$j = \boxed{5} \rightarrow j <= 4$
↓
False → Sopln()

} inner loop मधून बाहेर पडणार
i increment होणार.

$\boxed{i = 3 \rightarrow i <= 4 \rightarrow \text{True}}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
Sop(3) → $\boxed{j++}$

1 1 1 1
2 2 2 2
3

$j = \boxed{2} \rightarrow j <= 4$
↓
True
Sop(3) → $\boxed{j++}$

1 1 1 1
2 2 2 2
3 3

$j = \boxed{3} \rightarrow j <= 4$
↓
True
Sop(3) → $\boxed{j++}$

1 1 1 1
2 2 2 2
3 3 3

$j = \boxed{4} \rightarrow j <= 4$
↓
True
Sop(3) → $\boxed{j++}$

1 1 1 1
2 2 2 2
3 3 3 3

$j = \boxed{5} \rightarrow j <= 4$
↓
False → sop(n);

} inner loop च्या बाहेर येणार
i ~~ते~~ increment करणार.

$\boxed{i = 4 \rightarrow i <= 4 \rightarrow True}$

$j = \boxed{1} \rightarrow j <= 4$
↓
True
sop(4) → $\boxed{j++}$

1	1	1	1
2	2	2	2
3	3	3	3
4			

$j = \boxed{2} \rightarrow j <= 4$
↓
True
sop(4) → $\boxed{j++}$

1	1	1	1
2	2	2	2
3	3	3	3
4	4		

$j = \boxed{3} \rightarrow j <= 4$
↓
True
sop(4) → $\boxed{j++}$

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	

$j = \boxed{4} \rightarrow j <= 4$
↓
True
sop(4) → $\boxed{j++}$

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

$j = \boxed{5} \rightarrow j <= 4$
↓
$\boxed{False}$ → sop(n)

} inner loop च्या बाहेर पडणार
i increment होणार.

$\boxed{i = 5 \rightarrow i <= 4 \rightarrow False}$

outer loop ची condition False झाली त्यामुळे loop च्या बाहेर पडणार.

$\boxed{\text{ENJOYED}}$